Decisions

- Retain two patterned SAM layers and evaluate pressure bonding
- Retain the 2029 baseline while exploring 2028 acceleration
- Specify flexible dosing and zoning while protecting confidentiality

Open questions

- Validate screen, airflow, and drum design for registered zoning
- Validate winding methods that preserve resilience and third intake
- Define continuous-process controls for density, distribution, and contamination
- Confirm ND33 space and integration feasibility

Agenda

- Introductions and lunch (120 min)
- COCO V6 equipment discussion (255 min)

Meeting notes

Pattern Design

- Claire described two leading pattern designs from the pilot machine line: the leaf pattern and the Equipment pattern, evaluated using flat testing and U-shaped intake testing
- Pattern-to-pattern alignment remained an unresolved follow-up question, alongside open questions about fan size, fiber porosity, and pressure loss
- Claire said parallel jobs needed to use the same size, with the design requiring pad-to-pad alignment without white cutoffs
- Participants discussed whether the embossing should remain consistent across all 11 sizes, but no final decision was stated
- Claire explained that the two-pattern, two-layer direction was driven by business objectives and had been aligned with the China leadership team; different options and samples had been shown to the marketing and leadership teams

Layer Configuration

- The trial using a leaf-pattern top layer with a plain bottom layer did not perform well, so the design would continue using two layers with special pattern designs
- Participants discussed that the two-layer channel pattern performed better than the one-layer pattern in the cited testing, with increased void volume proposed as a possible explanation. The reason remained under discussion
- For phase one, Claire said the team was focusing on a five-layer material structure using 100% SAP, specifically SAP one and SAP two

- Claire explained that the two SAP layers were intended to serve different functions: the skin-side layer required conditioning and distribution, while the second layer supported rapid absorption; using the same SAP amount on both sides could sacrifice absorbency
- Claire explained that the line-side and polish-side SAP layers were expected to serve different functions: the line-side layer required high AUL, high GBT, and larger particle-size distribution for intake, while the polish-side layer required high CR, high Vortex, and relatively smaller particle-size distribution to reduce hydrogen-block issues
- Claire described an ultimate SAP density split of 70% versus 30%, with approximately 30% of the high-AUL SAP placed in the top layer to balance absorbency levels
- Claire interpreted the proposed structure as using a higher basis weight of high-CRC SAM in the bottom layer and high-AUL SAM in the top layer, rather than the same basis weight in both SAM layers
- Claire explained that using two SAM delivery systems would enable different SAP inputs and coverage on each layer
- Claire said the process would use two SAP dosing systems plus an additional dosing system for functional granules, which could be placed in either the LSS or PSS for some chemicals
- Claire explained that front and middle sections of one product could use different SAP-head volumes, and that the required speed would be calculated from the relevant dimensions, RPM, and PLS relationship
- Claire described a two-layer formulation in which the first layer combined 2D, EDGE, and 3D components, while the second layer used functional fibers including BCTMP side-by-side and PVA
- Claire said the current approach used a tropical liner without ADL coverage and considered adding more tri-level fiber to increase performance
- Claire described a two-layer configuration in which one layer used approximately 10% of the discussed material and the other used up to 50%

Product Structure

- The structure was described as five layers, including two patterned SAM layers matched to each other and an uncut Aurelia layer; bonding was considered necessary to preserve integrity and prevent the SAM from swelling into the voids
- Claire said the initial focus was on using 100% SAP on the poly side, while patterning on the other side was being considered for a later phase
- Participants agreed that REL cutting was not needed for bonding, although cutting could be used later to enable flash-situation distribution
- Claire described a redesigned fiber structure using 2D and 3D binder components for bonding and strength, with hollow fibers intended to improve resilience by acting like springs

Commercialization Timeline

- Claire said the commercialization timeline remained 2029. China leadership wanted commercialization as soon as possible, but commercialization next year was not considered feasible; the team would try for 2028 without committing to it
- Claire said the intended product volume was 7,000 products in 2029, while the design speed would be higher because the motors could operate at higher speed; she also referenced 20% over-banking
- Claire said commercialization was being prevented by the need for a second draw

Process Direction

- Claire said the team wanted to decide by the middle of November whether to retrofit the existing ND33 or install a new pilot machine in the Beijing machine bay
- Claire said the team might accept compromises on the data target while focusing on the larger pants-boring requirement, and that the 900 PPM target could be modified after a deeper discussion by mid-November

Process Design

- Claire said the process objective was to deliver flexible SAP density ADF placement regardless of the absorbent-core pattern or shape
- Claire said the process should support different SAP densities across zones and allow multiple absorbency sections when the material could be segmented accordingly
- Claire clarified that patterns should be treated as a special case for areas with specific SAP density requirements; technically, the process should support any designed SAP density at any desired location
- Claire said the channel pattern needed clear basic geometry with low SAP density, accurate pattern alignment and registration, and control of pathway deviations at high speed
- Claire described 11 pattern variations using different MD lens and CD width configurations, with the designs intended to maintain the same overall density target as the global V6
- Participants noted that the equipment discussion was separate from the two-SAM development work because the equipment already had two SAM feeders; the feeder assignments and a maximum target of 260 still needed to be specified
- Claire clarified that a defined total SAP add-on for each target did not prevent zoning or using different SAP densities across layers
- Claire said the pilot line was intended to provide more flexibility for testing configurations rather than function as a production machine
- Claire said the machine could be built to support the proposed two-system configuration, while the detailed implementation remained an engineering solution
- Claire said engineering needed to determine airflow velocity, the appropriate lens and height for mixing SAP and fluff, and how SAP particle-size distribution affected the forming process
- Claire said the forming fan would generate pressure and vacuum, with forming-bed RPM and controlled open air affecting SAP density in specific pad areas

- Claire explained that battery volume affected acid density, and said the 82.5 option had been rejected because it was not flexible enough to support different MD and CD configurations
- Participants discussed a risk that a single SAP layer could block airflow through the screen when vacuum was applied only from the bottom
- Participants discussed a layered screen design with vacuum applied from the bottom and sides of miniature sand pockets as a possible way to address the airflow limitation
- Claire identified the relationship between job, drum, and OD sizes as a key process risk: different sizes could change the nip gap and alignment, while larger jobs could increase thermal deformation and COV
- Participants discussed that the current adhesive allocation was average, while adding adhesive at specific positions would require a way to adjust the adhesive amount by location
- Claire said a trial had been completed the previous weekend and that the drums had been cleaned four times; applying adhesive to the drum surface was easy, but it created web-handling problems
- Claire said the proposed job size was only an idea and still needed verification by the engineering team
- Claire described a two-job process for the 11 different sizes under consideration
- Claire said the channel was designed at 8 millimeters for both the main and side areas, with a 3-millimeter embossing planned in the middle of the 8-millimeter channel
- Participants discussed testing different open-air settings to determine whether they produced distinct SAP densities, including whether areas with 10% and 15% open air would perform differently
- Claire discussed verifying whether different SAP densities could be produced using the same hole diameter, or alternatively by varying the hole diameters
- Claire described a third design path using a concave 3D screen, where gravity combined with open-air pressure was expected to help retain more SAP; the detailed design was still pending
- Claire discussed a fourth path using a flat screen with different open-air settings and a block approximately 3 millimeters high around the channel, with embossing at the same time being considered
- Participants discussed varying the SAM feed rate across different zones within one product rather than maintaining a continuous feed rate
- Participants discussed coordinating SAM feed timing with product or part position, including the latency involved when the feed speed is turned on or slowed down
- Participants discussed whether vacuum could move the SAM and whether controlling the material’ s position or bounce-back could reduce the timing needed to receive the SAP
- Claire said the SGCH-related process used around 10% fluff and did not require stuffing, while the alternative process required stuffing
- Claire identified maintaining the SAM placement as a major challenge and discussed how distribution and profile constraints could affect material retention

- Claire said phase one would focus on the current design challenges, while profiling work would be addressed in phase two
- Participants discussed core winding and surface winding as possible approaches for controlling tension, with surface winding described as providing more consistent tension
- Claire clarified that the proposed guidelines were intended for process evaluation rather than product-performance checks
- Claire described the work as an initial process evaluation using 3D training parts to test patterns and analyze the results
- Claire raised engineering concerns about equipment requirements, available space, structural changes, and establishing a sustainable process before proceeding
- Claire said the team had tried the alternative long-shoot method, but no clear preference between the methods was established

Absorbency Development

- For phase two, Claire said the team was considering adding a minimum fluff layer to the SCP2 absorbency layer to enable capillary absorbency, with a target of around 30 GSM or roughly 1 gram of material
- Claire said phase two was expected to consider adding fluff on the poly side to support absorbency, while the current process layered materials directly from layer to layer
- Claire described the ideal position as combining fluff with SAP, with the possibility of increasing the fluff proportion in some countries based on absorbency requirements
- A trial planned for next month at KCPX was expected to use two different types of SAM across multiple configurations to evaluate the resulting structures
- Claire said the process discussion involved evaluating a 10% fluff addition, whereas prior additions had been at least 30%
- In a comparison with other commercial materials, the tested IO material was reported to contain 62% SAP versus approximately 20% in the other materials; participants were still evaluating whether the higher SAP content could occupy the RDL void volume when the particles swell
- Claire said the phase-two work was still waiting for test results from the QA five-one pipeline

Equipment Procurement

- Claire clarified that purchasing from HCH would provide equipment for a pilot line, not technical support, so the team would handle development internally
- Claire said equipment selection depended on the production inputs, and that some input information would need to be shared with HCH
- Participants discussed defining design criteria so vendors could demonstrate the equipment’ s ability to produce the stated requirements; equipment that met the criteria could then be accepted
- Claire said the team did not yet know whether the proposed nozzle existed as available equipment, and that the issue could be critical because no machine was currently available

- Claire said the system specification still needed to define the zoning, static pressure, and airflow range; a vendor software tool could provide results from airflow inputs such as open-air CFM
- A three-forming-job system had been proposed to reduce the SAP load on each job, but the available space did not support that configuration
- Claire said the team was checking information from vendors and considering whether different OEM factors could require different operating approaches
- Claire said the team needed to compile the solution’ s capex and lead-time information so leadership could make the final decision on whether to proceed with the direction or change it

Confidential Design Information

- Claire said confidential design details, including density, MD-direction separation, urea accumulation, and the selection of top- and bottom-layer materials, would not be shared with HCH

Vendor Negotiation

- For the meeting with HCH, Claire proposed determining what HCH already knew and what had been shared with HSA, limiting information disclosed by the team, and asking HCH to distinguish proven equipment capabilities from developmental work and unknown learning risks

Equipment Design

- Claire said drum runout needed to be evaluated at approximately 150 RPM, noting that heat-related axis deformation could worsen the runout
- Claire said the two-drum schematic shown was a conceptual theory rather than the actual machine-data layout, and questioned whether using one drum to control the process made sense
- Claire discussed 11 sizes with different circumferences and widths, including widths of 100, 110, and 120, and said the machine evaluation needed to consider the smallest and largest sizes
- The required SAM feed and vacuum arrangement remained dependent on the gap between the SAP drop point and the receiving point, which participants said was not yet known
- Participants noted that forming-drum and forming-chamber design would need to account for airflow, mixing, and changes occurring over a defined distance
- Claire discussed whether a forming-chamber or wheel design could maintain the same capacity with an inverter motor and whether an angled configuration could improve material drop and forming performance
- Participants discussed the need to reduce tension during the winding process and asked for engineering guidance on how to instruct the vendor
- Claire described a second-shell process that used a drone and indicated that the package might need to be changed, although the fixed vacuum limited how much could be modified

- Claire said the OEM process used a chamber and a changing pipeline or shoot, and that the chamber was helpful for material distribution, although the team did not yet know exactly what occurred inside it
- Participants discussed the need to determine how the chamber behaved during continuous operation and how to detect material density, distribution, contamination, web variation, and quality at high speed; speeds of 1,000 and 1,200 were referenced
- Claire identified drum diameter, static pressure, and sand velocity as process variables, but the appropriate settings remained unresolved
- Participants discussed adding another form or module, which would require additional space for fans and potentially relocating equipment; the layout remained unresolved
- Claire said the raw-material channel might require a second or larger vacuum system, with loading or repeating systems also considered
- Participants said the available space supported only two rows at one time with no spare capacity, while building-wall and aisle constraints affected bulk placement

Pilot Trial Design

- Claire said a three-section trial was planned for the campaign pilot lab in Q1; if the three-section design worked, it could be extended to five sections
- Claire said a pilot or machine line could potentially be built to verify the proposed structure for YP or another country and evaluate a hybrid structure, but this remained under consideration
- Claire described a test layout using two drops that was separate from the current machine configuration

Material Testing

- Testing indicated approximately 7% in one reported performance measure, while folding capacity was nearly doubled; the exact measure associated with the 7% result was unclear
- Claire reported that initial performance improved on the first and second attempts but was slightly worse on the third attempt; the explanation remained an assumption based on the pilot work
- Claire referred to the sample obtained during the second trial as part of the ongoing evaluation
- Claire described testing samples from the existing shelf supply against OEM samples to compare their reactions and refine the issue identified in the first trial
- Claire reported that the first trial used one layer and that the available data was incomplete; later trials changed process and material variables for comparison
- Claire said performance comparisons were limited because the material used was not commercial material

Material Development

- Claire said PBA fiber had been added because the PPP fiber did not provide sufficient physical capability
- The tested sample material had been produced by the fiber producer during its pilot, and participants discussed moving to a commercial manufacturer for the next stage
- Claire said the team had gained experience with different fibers through Danny, but questioned whether that experience was sufficient for future tracking work
- Claire reported that adding IEL before the draw appeared to benefit SAM capture, so the team used IEL together with PSM in the trial

Process Development

- Claire described the third trial as incomplete and said it included additional process or material changes applied continuously during the trial
- Claire said the current work was at step two and that an FTS request had already been made

Layout Design

- Claire said the layout being discussed was a draft rather than the new layout

Follow-up tasks

Task	Assigned to	Due date	Bucket
Provide local marketing guidance on pattern-design principles, including the required pattern dimensions and the combination of vertical and branch patterns (Claire)			
Test pressure bonding by the end of this year as an alternative to ultrasonic bonding (Claire)			

Remind the engineers about the critical design consideration for allocating adhesive density by position (Claire)			